21 The MVC and Razor Pages filter pipeline

This chapter covers

- The filter pipeline and how it differs from middleware
- The different types of filters
- Filter ordering

Part 3 of this book has covered the Model-View-Controller (MVC) and Razor Pages frameworks of ASP.NET Core in some detail. You learned how routing is used to select a Razor Page or action to execute. You also saw model binding, validation, and how to generate a response by returning an `IActionResult` from your actions and page handlers. In this chapter I'm going to head deeper into the MVC/Razor Pages frameworks and look at the *filter pipeline*, sometimes called the *action invocation pipeline*, which is analogous to the minimal API endpoint filter pipeline you learned about in chapter 5.

MVC and Razor Pages use several built-in filters to handle cross-cutting concerns, such as authorization (controlling which users can access which action methods and pages in your application). Any application that has the concept of users will use authorization filters as a minimum, but filters are much more powerful than this single use case. In sections 21.1 and 21.2 you'll learn about all the different types of filters and how they combine to create the MVC filter pipeline for a request that reaches the MVC or Razor Pages framework.

Think of the MVC filter pipeline as a mini middleware pipeline running inside the MVC and Razor Pages frameworks, like the minimal API